

TUTORIAL SHEET - 2

(DIGITAL CIRCUITS AND SYSTEMS)

1. A Boolean function $F(A,B,C,D) = \prod M(1,5,12,15)$ is to be implemented using :

(a) 8x1 MUX (A is MSB). The inputs B, C & D are connected to the select input S_2 S_1 S_0 of the MUX respectively. Design circuit using 8x1 multiplexer.

(b) 4x1 MUX (A is MSB). The inputs B & C are connected to the select input S_1 & S_0 of the MUX respectively. Design circuit using 4x1 multiplexer.

2. Use a 4-to-1 multiplexer and a minimum number of external gates to realize the function :

$$F(W, X, Y, Z) = \sum m(3, 4, 5, 7, 10, 14) + \sum d(1, 6, 15).$$

The inputs are only available uncomplemented. (LSB as selection lines)

3. A combinational circuit is specified by the following three Boolean functions:

$$F_1(A, B, C) = \sum (2, 4, 7)$$

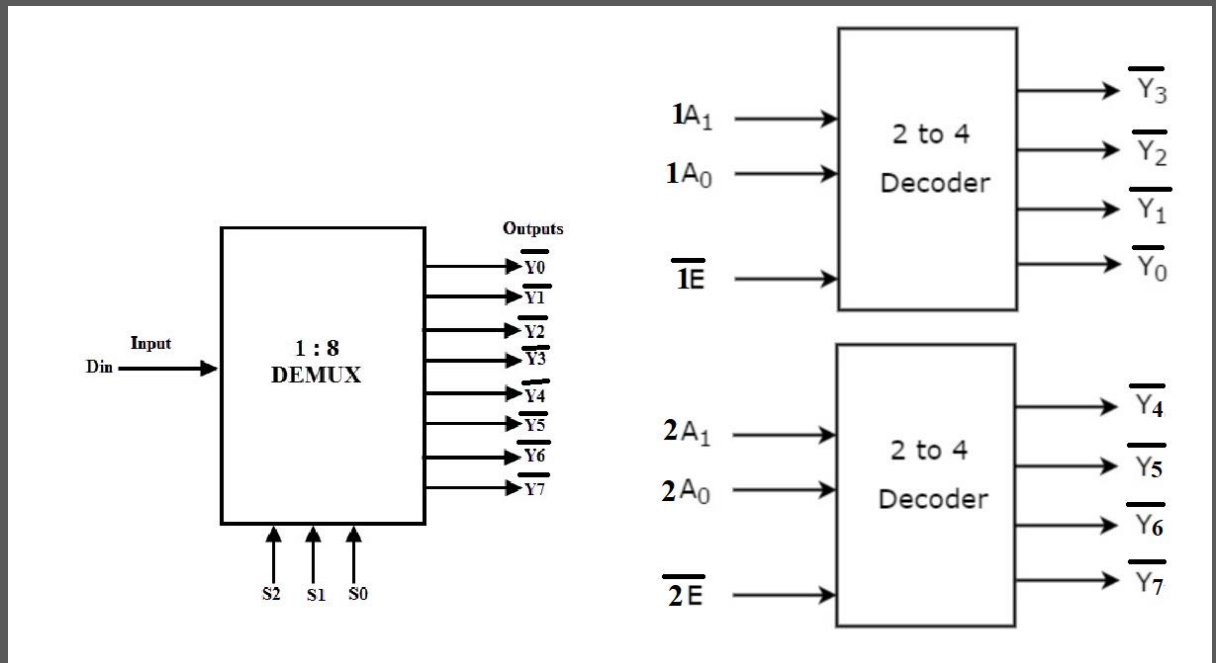
$$F_2(A, B, C) = \sum (0, 3)$$

$$F_3(A, B, C) = \sum (0, 2, 3, 4, 7)$$

Implement the circuit with a decoder and suitable gates.

4. Realize a BCD to excess-3 code converter using a 4-to-10 decoder with active low outputs and a minimum number of gates.

5. A 1:8 demultiplexer with data input D_{in} , address inputs is to be designed $S_2S_1S_0$ (with S_0 as LSB) and Y_0' to Y_7' as the eight demultiplexer outputs, is to be designed using two 2:4 decoders (with enable input E' and address inputs A_0 & A_1)



6. Design a 1:7 line demultiplexer with as many 1:2 line demultiplexers as are needed.